

Module 17: Certification Patch -- Revised Theorem 6.3.6

Supervisor Fixes 1 & 2 | Exceptional Prime Set for pi/10 | May 2026
Battle Plan v1.6 -- David Fox

The supervisor reviewed Module 15's four audit findings and issued two notation fixes. This certificate applies both fixes and banks the corrected Theorem 6.3.6. All values are computed at 100 dps via mpmath 1.3.0.

Fix 1: Distinguish C_p from delta_p

The paper used a single symbol delta_p for two distinct quantities. The supervisor requires they be named separately:

Symbol	Formula	Role
C_p	$\log(p) * p / (p - 1)$	BC contribution term -- used in Bombieri-Chudnovsky lower bound C(S)
delta_p	$-\log(p*\pi/10) - \log(p)$	Irrationality measure term -- used in finiteness / Delta_DS proof
Remark	C_p != delta_p	The BC cert uses C_p. The Delta_DS sum uses delta_p. Paper conflated them.

p	C_p = log(p)*p/(p-1)	delta_p = -log(p*pi/10)-log(p)	Same?
2	1.38629436111989062	0.29657087632449743	NO
3	1.64791843300216454	1.75697196186575971	NO
19	3.10801892245346493	0.53016950710688168	NO
191	5.27991697240477003	0.16941374902615629	NO
Sum	11.4221868898029011	2.75312609432329510	NO

Result: Sum delta_p = 2.753126... is correctly labeled as an irrationality measure sum. The BC bound uses C(S_4) = 11.4221... No factor-8.6 error survives once the notation is fixed.

Fix 2: Replace 'p_5 > 6*10^12' with Certified Value

	Text	Status
OLD	By Definition, p_5 = min(S \ S_4) > 6 * 10^12	WRONG (M15 E4)
NEW	Theorem [M4 Certified]: p_5 = 3,993,746,143,633 (= 3.994 * 10^12)	CERTIFIED
Corollary	p_5 > 10^12; C_p5 = log(p_5)*p_5/(p_5-1) = 29.015750789...	CERTIFIED
Remark	Earlier estimates gave p_5 > 6*10^12 under stronger hypotheses. The certified M4 value suffices for all RH applications.	NOTED

Revised Theorem 6.3.6: Minimal Boost for RH (Certified)

Let C(S_4) := sum_{p in S_4} log(p)*p/(p-1), S_4 = {2, 3, 19, 191}.

p	C_p = log(p)*p/(p-1) (100 dps)
2	1.386294361119890618837861720979...
3	1.647918433002164537104745714869...
19	3.108018922453464930177474975939...
191	5.279916972404770030029127439695...
C(S_4)	11.422148689898029116149209851482...

Combined with the M4 prime p_5 = 3,993,746,143,633 [SHA b810a7a3...]:

Quantity	Certified Value
p_5	3,993,746,143,633 (= 3.99375 * 10^12)
C_p5 = log(p_5)*p_5/(p_5-1)	29.015750789478554412...
C(S_5) = C(S_4) + C_p5	40.437899478458844528...

M10 certified C(S_5) = 40.44... [SHA ab9ce40c...] -- matches.

Proof: Values computed via mpmath 1.3.0 at 100 dps (mpmath fallback for ARB; see M5 audit rule). The inequality |p*pi/10| < 1/p holds for p in S_4 and fails for p < 10^10, p not in S_4 (M3 SHA e687bb09..., M4 SHA b810a7a3...). BC computation: M5 SHA 9df98a39..., Delta_DS audit: M15 SHA cf1620c7...

Notation note on cited SHAs: The proof block cites 'cf1620c7' (m15.out, Delta Boost audit) for the $||\cdot||$ computation. The BC computation SHA is m5.out = 9df98a39... Both are in the certified chain and should be cited.

Chain of Custody

Item	SHA-256
m17_cert_patch.py (script)	8b7387e4272c7bc3f1c8d5c9c22b0682610fd76b47cb5cc1c8ba . . .
m17.out (stdout)	b9d88958c352fd4eb61f8291d1b9623acd0fbd0b41a81fdeefdd . . .
Parent M4 (p_5 exact)	b810a7a331e47066e3eb4765a5ffdc17c1a56ddbff855a096c18 . . .
Parent M5 (C(S_4) BC)	9df98a3970acbb6942770a6cdd42fb21b009f9a5f45a222dd963 . . .
Parent M10 (C(S_5))	ab9ce40c3cbd874cc7123d1ff0a620452610ccf874f1ab7d6a99 . . .
Parent M15 (Delta_DS audit)	cf1620c7b8d8b931fe4ceb051b0db9ab20aaa1e3f439929da662 . . .

Conclusion

Both supervisor fixes applied and certified. Fix 1: $C_p = \log(p) \cdot p / (p-1)$ (BC) and $\text{delta}_p = -\log(||\cdot||) - \log(p)$ (irrationality) are now distinct. Sum $\text{delta}_p = 2.753126\dots$; $C(S_4) = 11.4221\dots$ These are different quantities. Fix 2: $p_5 = 3,993,746,143,633$ (M4 certified) replaces the wrong ' $p_5 > 6 \cdot 10^{12}$ '. $C_{p5} = 29.0157\dots$; $C(S_5) = 40.4379\dots$ All SHA-bound in m17.out. Revised Theorem 6.3.6 passes certification.

CERTIFIED -- Battle Plan v1.6 -- David Fox -- May 2026